

Cline’s formula fails for one-sided Drazin inverses

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Source and answer

Yan [1] proves that the nonzero parts of the one-sided Drazin spectra of ac and ca agree, and asks in Question 3.10 whether the missing zero point also agrees—equivalently, whether Cline’s formula holds for one-sided Drazin inverses in rings and one-sided generalized Drazin inverses in Banach algebras.

By Corollary 3.9, we immediately obtain the following spectral identity for one-sided (generalized) Drazin spectrum.

Corollary 3.9. *Let $a, c \in \mathcal{A}$. Then*

$$\sigma_*(ac) \setminus \{0\} = \sigma_*(ca) \setminus \{0\},$$

where $*$ = {ld, rd, lg, rg, lgd, rgd}.

It is also interesting to consider whether the Cline formula holds for one-sided (generalized) Drazin inverses in Banach algebras, namely the relation between the zero points of the one-sided (generalized) Drazin spectrum of element products ac and ca in Banach algebras. However, we conjecture that Cline formula may not hold for one-sided (generalized) Drazin inverses.

Question 3.10. *Does the Cline formula hold for one-sided (resp. generalized) Drazin inverses in rings (resp. Banach algebras)?*

We now present Cline formula for one-sided Drazin inverses under certain additional conditions. A comparable result concerning one-sided generalized Drazin inverses can be derived through analogous computations.

The answer is no, simultaneously for the left and right versions and for both Drazin and generalized Drazin invertibility. The counterexamples lie in the unital Banach algebra $\mathcal{B}(\ell^2)$, so they also answer the ring version.

Definitions

An element T of a unital Banach algebra is *left generalized Drazin invertible* if some X satisfies

$$TXT = XT^2, \quad X^2T = X, \quad TXT - T \text{ is quasinilpotent.}$$

Replacing “quasinilpotent” by “nilpotent” gives left Drazin invertibility. The right definitions reverse the displayed identities. Every left invertible element is left Drazin invertible (of index zero), and similarly on the right.

For $T \in \mathcal{B}(\mathcal{H})$, write

$$H_0(T) = \left\{ x \in \mathcal{H} : \lim_{m \rightarrow \infty} \|T^m x\|^{1/m} = 0 \right\}.$$

Proof intuition

Split the coordinates of ℓ^2 into even and odd positions. Let D embed the whole space isometrically into the even coordinates. Let C be the identity on those coordinates but shrink the n th odd coordinate by $1/n$. In the order CD , the shrinking part is never seen, so $CD = D$ is left invertible. In the reverse order DC , every vector first experiences C and is then pushed repeatedly through isometric even-coordinate embeddings. Thus no nonzero vector decays quasinilpotently, although the odd basis vectors are sent to vectors of norm $1/n$. A short idempotent argument shows that a left generalized Drazin inverse would then have to be an actual left inverse, which is impossible.

An algebraic obstruction

Lemma 1. *If $T \in \mathcal{B}(\mathcal{H})$ is left generalized Drazin invertible and $H_0(T) = \{0\}$, then T is left invertible.*

Proof. Let X be a left generalized Drazin inverse and put $Q = XT$. The defining identities give

$$Q^2 = XTXT = X^2T^2 = XT = Q, \quad TQ = TXT = XT^2 = QT.$$

Moreover, $T - TQ = T(1 - Q)$ is quasinilpotent. Since T commutes with Q , for every $z \in \text{Ran}(1 - Q)$ and every $m \geq 1$,

$$T^m z = [T(1 - Q)]^m z.$$

The spectral-radius formula therefore implies $\text{Ran}(1 - Q) \subseteq H_0(T) = \{0\}$. Hence $Q = 1$, so $XT = 1$ and T is left invertible. $\square$

The counterexample

Theorem 1. *There exist $a, c \in \mathcal{B}(\ell^2(\mathbb{N}))$ such that ac is left Drazin invertible but ca is not left generalized Drazin invertible. There also exist such a pair with “left” replaced everywhere by “right”. Consequently, Cline’s formula fails for every one-sided class asked about in Question 3.10.*

Proof. Let $(e_n)_{n \geq 1}$ be the standard basis of $\mathcal{H} = \ell^2(\mathbb{N})$ and define $D, C \in \mathcal{B}(\mathcal{H})$ by

$$De_n = e_{2n}, \quad Ce_{2n} = e_{2n}, \quad Ce_{2n-1} = \frac{1}{n}e_{2n-1} \quad (n \geq 1).$$

The operator D is an isometry, $D^*D = 1$, and C acts as the identity on $\text{Ran } D$. Thus

$$CD = D.$$

In particular, CD is left invertible, with left inverse D^* , and hence is left Drazin invertible.

Set $T = DC$. From $CD = D$, induction gives

$$T^m = (DC)^m = D^m C \quad (m \geq 1).$$

Because every D^m is an isometry,

$$\|T^m x\| = \|Cx\| \quad (x \in \mathcal{H}, m \geq 1).$$

The diagonal operator C is injective. Hence for $x \neq 0$,

$$\lim_{m \rightarrow \infty} \|T^m x\|^{1/m} = \lim_{m \rightarrow \infty} \|Cx\|^{1/m} = 1,$$

and therefore $H_0(T) = \{0\}$.

On the other hand,

$$\|Te_{2n-1}\| = \frac{1}{n} \longrightarrow 0.$$

Thus T is not bounded below. It cannot have a bounded left inverse, because $XT = 1$ would imply $\|x\| \leq \|X\| \|Tx\|$ for every x . Lemma 1 now shows that $T = DC$ is not left generalized Drazin invertible. Taking $a = C$ and $c = D$ proves the left assertion, and also disproves the weaker left-Drazin closure.

For the right assertion, take $a = D^*$ and $c = C = C^*$. Then

$$ac = D^*C = (CD)^* = D^*, \quad ca = CD^* = (DC)^*.$$

The coisometry D^* is right invertible, with right inverse D . If $(DC)^*$ were right generalized Drazin invertible, taking adjoints in its three defining identities would make DC left generalized Drazin invertible; adjoints preserve quasinilpotence in $\mathcal{B}(\mathcal{H})$. This contradicts the first half of the proof. Hence the right versions fail as well. $\square$

Later-literature audit and scope

A later paper of Kolundžija and Mosić [2], published online on 21 October 2025, studies conditional Cline formulas for the same one-sided generalized inverses. Its Example 3.1 is intended to show that a commutation condition cannot be omitted and asserts a left generalized-Drazin failure. That particular example does not establish the assertion. In its notation, the reversed shift satisfies

$$BAe_1 = 0, \quad BAe_n = e_{n+1} \quad (n \geq 2).$$

Thus the reducing decomposition

$$\ell^2 = \text{span}\{e_1\} \oplus \overline{\text{span}}\{e_n : n \geq 2\}$$

writes BA as $0 \oplus S$, where S is a unilateral isometry. Hence BA is left Drazin invertible. The other diagonal block in their displayed operator is quasinilpotent, hence left generalized Drazin invertible with inverse zero; their whole reversed product is therefore left generalized Drazin invertible. A separate proof-gap record accompanies this packet.

The construction in Theorem 1 repairs that obstruction and is self-contained. Searches through 2026-08-12 of the four run indexes, the exact question and title, citations, and targeted one-sided Cline-formula literature found no other valid unconditional counterexample. This is a bounded novelty check, not an exhaustive survey of all non-indexed literature.

The theorem resolves the existence/equivalence part of Question 3.10 in the negative for both orientations and both notions. It does not characterize the additional hypotheses under which a one-sided Cline formula is valid.

Human review recommendation

Review as a likely-valid full counterexample. The decisive checks are the identity $(DC)^m = D^m C$, the conclusion $H_0(DC) = \{0\}$, and Lemma 1. The later-paper proof-gap note should be reviewed independently from the validity of the new construction.

References

- [1] Kai Yan, *On the spectral identities and fundamental properties of one-sided Drazin inverses in Banach algebras*, arXiv:2504.18995 (2025), Question 3.10.
- [2] Milica Z. Kolundžija and Dijana Mosić, *Properties of one-sided generalized Drazin inverses in Banach algebras*, *Advances in Operator Theory* **11**, Article 2 (2026), published online 21 October 2025, doi:10.1007/s43036-025-00479-1.